

AWS CHEAT SHEET

Amazon Web Services — Complete Quick Reference Guide

What is AWS?

- ▶ Amazon Web Services — world's leading cloud platform with 200+ services.
- ▶ Offers compute, storage, databases, AI/ML, networking & more.
- ▶ Pay-as-you-go pricing model — no upfront cost.
- ▶ Available in 30+ geographic Regions with 90+ Availability Zones (AZs).

EC2 — Elastic Compute Cloud

- ▶ Virtual servers in the cloud (choose OS, CPU, RAM, storage).
- ▶ Instance types: General (t3), Compute (c5), Memory (r5), GPU (p3).
- ▶ Pricing: On-Demand | Reserved (up to 75% off) | Spot (up to 90% off).
- ▶ AMI (Amazon Machine Image) = template to launch EC2 instances.
- ▶ Security Groups = virtual firewall for inbound/outbound traffic.
- ▶ Elastic IP = static public IP address for your instance.

S3 — Simple Storage Service

- ▶ Object storage for any file type — unlimited capacity.
- ▶ Objects stored in Buckets; each object can be up to 5TB.
- ▶ Storage classes: Standard | Intelligent-Tiering | Glacier (archive).
- ▶ Versioning — keep multiple versions of an object.
- ▶ S3 Static Website Hosting — host HTML/CSS/JS websites easily.
- ▶ ACL & Bucket Policies control access; Block Public Access by default.

IAM — Identity & Access Management

- ▶ Controls WHO can do WHAT in AWS — free service.
- ▶ Users = individual accounts | Groups = collection of users.
- ▶ Roles = temporary permissions (used by EC2, Lambda, etc.).

- ▶ Policies = JSON documents defining permissions (Allow/Deny).
- ▶ Principle of Least Privilege — give only the access needed.
- ▶ MFA (Multi-Factor Authentication) — always enable for root account.
- ▶ Root account = full access; never use for daily tasks.



Databases

- ▶ RDS — Managed relational DB (MySQL, PostgreSQL, Oracle, SQL Server).
- ▶ Aurora — AWS-native DB; 5x faster than MySQL, 3x faster than PostgreSQL.
- ▶ DynamoDB — Serverless NoSQL, single-digit millisecond latency.
- ▶ ElastiCache — In-memory caching (Redis / Memcached).
- ▶ Redshift — Data warehouse for analytics & BI workloads.
- ▶ Multi-AZ = high availability | Read Replicas = read performance.



Networking & VPC

- ▶ VPC (Virtual Private Cloud) — isolated network environment in AWS.
- ▶ Subnets: Public (internet-facing) | Private (internal only).
- ▶ Internet Gateway — allows VPC to communicate with internet.
- ▶ NAT Gateway — lets private subnet instances access internet (outbound only).
- ▶ Route Tables — control traffic routing within VPC.
- ▶ Security Groups (stateful) vs NACLs (stateless) — both control traffic.
- ▶ VPC Peering — connect two VPCs privately.
- ▶ Direct Connect — dedicated private connection from on-premise to AWS.



Lambda — Serverless Computing

- ▶ Run code without managing servers — pay per invocation.
- ▶ Supports Python, Node.js, Java, Go, Ruby, .NET, and more.
- ▶ Max execution time: 15 minutes | Max memory: 10 GB.
- ▶ Triggers: API Gateway, S3, DynamoDB, EventBridge, SQS, etc.
- ▶ Free tier: 1 million requests/month + 400,000 GB-seconds compute.



High Availability & Scaling

- ▶ Auto Scaling — automatically add/remove EC2 instances based on demand.

- ▶ **ELB (Elastic Load Balancer): ALB (HTTP/HTTPS) | NLB (TCP/UDP) | CLB (legacy).**
- ▶ **ALB — Layer 7, supports path/host-based routing.**
- ▶ **NLB — Layer 4, ultra-low latency, static IP support.**
- ▶ **Target Groups — route requests to registered EC2, Lambda, or IPs.**
- ▶ **Health Checks — ELB only routes to healthy instances.**

Containers & Kubernetes

- ▶ **ECS (Elastic Container Service) — run Docker containers on AWS.**
- ▶ **EKS (Elastic Kubernetes Service) — managed Kubernetes cluster.**
- ▶ **Fargate — serverless compute for containers (no EC2 management).**
- ▶ **ECR (Elastic Container Registry) — store & manage Docker images.**
- ▶ **ECS vs EKS: ECS is simpler; EKS is more flexible for Kubernetes users.**

Messaging & Integration

- ▶ **SQS (Simple Queue Service) — decouple apps with message queues.**
- ▶ **SNS (Simple Notification Service) — pub/sub messaging & push notifications.**
- ▶ **SQS Standard (at-least-once) vs FIFO (exactly-once, ordered).**
- ▶ **EventBridge — event bus to connect AWS services & SaaS apps.**
- ▶ **Kinesis — real-time streaming data (logs, IoT, clickstreams).**
- ▶ **SNS → SQS fan-out pattern = common architecture for decoupling.**

Content Delivery & DNS

- ▶ **CloudFront — CDN with 400+ edge locations for low-latency delivery.**
- ▶ **Route 53 — scalable DNS service + domain registration.**
- ▶ **Routing policies: Simple | Weighted | Latency | Failover | Geolocation.**
- ▶ **CloudFront Origins: S3, EC2, ALB, or custom HTTP servers.**
- ▶ **OAI (Origin Access Identity) — restrict S3 access to CloudFront only.**

Security Services

- ▶ **KMS (Key Management Service) — create & manage encryption keys.**
- ▶ **CloudTrail — logs all API calls in AWS (who did what & when).**
- ▶ **GuardDuty — ML-based threat detection (brute force, crypto mining).**

- ▶ **Shield** — DDoS protection (Standard = free | Advanced = paid).
- ▶ **WAF (Web App Firewall)** — protect against SQL injection, XSS, etc.
- ▶ **Secrets Manager** — store & rotate credentials, API keys, passwords.
- ▶ **Inspector** — automated security assessment for EC2 & containers.

Monitoring & Logging

- ▶ **CloudWatch** — metrics, logs, dashboards, and alarms.
- ▶ **CloudWatch Logs** — collect and store log files from services.
- ▶ **CloudWatch Alarms** — trigger actions (scale EC2, send SNS) on metrics.
- ▶ **X-Ray** — distributed tracing to debug and analyze microservices.
- ▶ **Config** — track AWS resource configurations and changes over time.
- ▶ **CloudTrail** — audit log of all AWS API activity.

Infrastructure as Code

- ▶ **CloudFormation** — provision AWS resources using JSON/YAML templates.
- ▶ **CDK (Cloud Development Kit)** — define infrastructure using Python, JS, Java, etc.
- ▶ **Elastic Beanstalk** — deploy apps without managing infrastructure.
- ▶ **OpsWorks** — configuration management using Chef/Puppet.
- ▶ **SSM (Systems Manager)** — manage EC2 at scale (patch, run commands).

AI / ML Services

- ▶ **SageMaker** — build, train & deploy ML models at scale.
- ▶ **Rekognition** — image/video analysis (face detection, labels).
- ▶ **Comprehend** — NLP: sentiment, entities, key phrases.
- ▶ **Lex** — build conversational chatbots (powers Alexa).
- ▶ **Polly** — text-to-speech service.
- ▶ **Translate** — real-time language translation.
- ▶ **Bedrock** — access foundation models (Claude, Llama, Titan) via API.

Cost Management

- ▶ **Free Tier**: 12 months free + always-free + short-term trials.
- ▶ **Cost Explorer** — visualize and analyze your AWS spending.

- ▶ **Budgets** — set spending/usage alerts and take automated actions.
- ▶ **Trusted Advisor** — recommendations on cost, performance, security.
- ▶ **Savings Plans** — flexible pricing vs Reserved Instances (up to 66% off).
- ▶ **Compute Optimizer** — right-size EC2, Lambda, ECS recommendations.

Global Infrastructure

- ▶ **Region** — geographic area with multiple AZs (e.g., us-east-1).
- ▶ **Availability Zone (AZ)** — one or more data centers in a region.
- ▶ **Edge Location** — CloudFront/Route 53 cache points (400+ worldwide).
- ▶ **Local Zones** — AWS infrastructure closer to end users for ultra-low latency.
- ▶ **Wavelength** — deploy apps at 5G network edge for mobile users.

AWS Well-Architected Framework

- ▶ **6 Pillars:** Operational Excellence | Security | Reliability | Performance | Cost | Sustainability.
- ▶ **Operational Excellence** — run & improve systems, automate changes.
- ▶ **Security** — protect data & systems with IAM, encryption, detection.
- ▶ **Reliability** — recover from failures; use Multi-AZ and backups.
- ▶ **Performance Efficiency** — use right resources for the workload.
- ▶ **Cost Optimization** — eliminate waste; use right pricing models.
- ▶ **Sustainability** — minimize environmental impact of workloads.

Quick Tips for Exams & Interviews

- ▶ **S3** = object storage | **EBS** = block storage | **EFS** = file storage.
- ▶ **RDS** = relational (SQL) | **DynamoDB** = NoSQL | **Redshift** = analytics.
- ▶ **CloudFront** caches content | **CloudTrail** logs API calls | **CloudWatch** monitors metrics.
- ▶ **SQS** = queue (pull) | **SNS** = pub-sub (push).
- ▶ **IAM** = who can do what | **KMS** = encryption keys | **Secrets Manager** = credentials.
- ▶ **Lambda** = serverless functions | **ECS/EKS** = containers | **EC2** = full VMs.
- ▶ **ALB** = Layer 7 (HTTP) | **NLB** = Layer 4 (TCP) | **Route 53** = DNS.
- ▶ **Always:** enable MFA, use least privilege, encrypt data, enable CloudTrail.

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